Hetchy Trail

Historical Game Mapping Brief

1. Recommended Historical Structure

Hetchy Trail should play as the construction of an interconnected system rather than as a literal wagon journey moving continuously west.

Although the aqueduct carries water from the Sierra Nevada toward San Francisco, construction did not occur in perfect geographical order. Crews worked on distant divisions simultaneously. Pulgas Tunnel and the first Bay Crossing pipeline were already carrying local Spring Valley water in the mid-1920s, while the Foothill, San Joaquin, and Coast Range divisions remained incomplete. The player's ultimate objective is therefore to connect all independently built sections into one continuous gravity system by October 1934.

Recommended campaign structure

1. Establish access to the High Sierra.
2. Build the dam and mountain headworks.
3. create construction power and penetrate the Sierra foothills.
4. descend through Moccasin and cross the western foothills.
5. lay the long San Joaquin Valley pipeline.
6. tunnel through the Coast Range.
7. cross Alameda Creek and San Francisco Bay.
8. connect the completed system at Pulgas.

The official SFPUC route proceeds through Hetch Hetchy, Early Intake, Mountain Tunnel, Priest Reservoir, Moccasin, Foothill Tunnel, the San Joaquin pipelines, Tesla Portal, Coast Range Tunnel, Alameda Creek, Irvington, the Bay Division pipelines, Pulgas, Crystal Springs, and ultimately San Francisco.

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2. Mile-Marker Convention

The mile ranges below are **game-design stationing**, not historical engineering survey stations.

Historical sources provide lengths for individual components, but they do not consistently present the entire system as one continuous 0–167 mile alignment. SFPUC describes Pulgas as the terminus of an aqueduct extending more than 160 miles from the Sierra. The commonly used 167-mile figure generally describes the broader source-to-San Francisco journey.

The proposed game miles therefore use rounded component lengths:

Phase	Approximate Game Miles	Historical Geography
1. High Sierra Access and Dam	0–12	Hetch Hetchy Valley to Early Intake
2. Mountain Tunnel and Moccasin	12–33	Early Intake to Priest and Moccasin
3. Western Foothills	33–49	Moccasin to Oakdale Portal
4. San Joaquin Valley	49–97	Oakdale Portal to Tesla Portal
5. Coast Range	97–126	Tesla Portal to Alameda Creek and Irvington
6. Bay and Peninsula	126–167	Irvington, Bay Crossing, Pulgas, Crystal Springs and city integration

The final phase compresses several connected Peninsula facilities into the remaining game mileage. Treat those numbers as navigational progression rather than exact construction stationing.

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3. Resource Translation

Use a five-point impact model in which **1 is a minor effect, 2 is substantial, and 3 is severe**.

Funds/Bonds

Represents cash on hand, approved bond authority, construction contracts, land acquisition, equipment and emergency appropriations.

Public Support

Represents voter confidence, willingness to approve bonds, confidence in the project and public enthusiasm for completing the water system.

Water Supply

Before the final connection, this should represent **system readiness and future capacity**, not water already delivered to San Francisco.

A completed dam, tunnel or pipeline increases readiness. Actual delivered Hetch Hetchy water should remain unavailable until the route becomes continuous.

Crew Wellbeing

Represents safety, fatigue, housing, sanitation, morale, retention and the ability to maintain a skilled workforce.

Time/Season

Represents construction seasons gained or lost. One point can equal approximately one season for balancing purposes.

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4. Route Map by Geographical Phase

Phase 1: High Sierra Access and Dam

Game miles 0–12 | Primarily 1914–1923

Outcome: The player must first create a functioning industrial corridor in terrain that initially lacks the roads, power, timber supply and transportation needed for a major dam.

Early Intake lies approximately 12 miles downstream from Hetch Hetchy. Before large-scale excavation could proceed, the city built roads, sawmills, camps and the Hetch Hetchy Railroad.

Principal physical challenges

- Remote mountain terrain
- Snowbound roads
- Transporting cement and heavy machinery
- Obtaining construction timber
- Diverting the Tuolumne River
- Excavating deep glacial material before reaching bedrock
- Maintaining continuous concrete placement through changing seasons

Historical encounters

1. **Build the access roads and sawmill.**
2. **Complete the Hetch Hetchy Railroad and use public excursions to strengthen bond support.**
3. **Divert the river, excavate through the glacial moraine and complete the first O'Shaughnessy Dam.**

Phase 2: Early Intake, Mountain Tunnel and Moccasin

Game miles 12–33 | Primarily 1917–1925

Outcome: The player converts the project from an isolated dam site into a powered mountain water-conveyance system.

The Mountain Tunnel extended roughly 19 miles from Early Intake toward Priest. It was excavated from portals, adits and shafts that created 12 working faces. Much of the tunnel ran approximately 1,000 feet beneath the surface.

Principal physical challenges

- Supplying electric power to remote drilling operations
- Maintaining accurate alignment from multiple tunnel headings
- Deep shaft access
- Hard granite and lined tunnel sections
- Protecting source-water quality at Priest Reservoir
- Controlling the 1,316-foot hydraulic descent to Moccasin

Historical encounters

1. **Build Early Intake Powerhouse before expanding tunnel work.**
2. **Coordinate 12 Mountain Tunnel excavation faces.**
3. **Complete Priest Reservoir, protect it from local runoff and harness the drop to Moccasin.**

Phase 3: Western Foothills

Game miles 33–49 | Primarily 1925–1929

Outcome: The player establishes several remote work camps and carries the aqueduct through the broken western Sierra foothills.

The Foothill Tunnel ran approximately 15.8 miles from the Moccasin area to Oakdale Portal. Ten excavation faces were supplied from portals, shafts and adits.

Principal physical challenges

- Establishing multiple self-contained construction camps
- Supplying water, electricity, roads and telephone service
- Coordinating city crews and private contractors
- Crossing the steep Red Mountain Bar canyon
- Completing work before the canyon was affected by Don Pedro Reservoir
- Balancing tunneling speed against workforce sustainability

Historical encounters

1. **Construct and supply six foothill work camps.**
2. **Build the Red Mountain Bar inverted siphon and cableway.**
3. **Decide how aggressively to pursue tunneling records.**

Phase 4: San Joaquin Valley Crossing

Game miles 49–97 | Primarily 1931–1932

Outcome: The player must transform a completed mountain tunnel system into a long-distance steel-pipeline operation across the valley floor.

San Joaquin Pipeline No. 1 extended approximately 47.5 miles from Oakdale Portal to Tesla Portal. The 56- to 72-inch steel pipeline was built in 1931–1932 and had an initial capacity of approximately 70 million gallons per day.

Principal physical challenges

- Procuring and assembling nearly 48 miles of large steel pipe
- Maintaining grade across long stretches of agricultural land
- Reserving room for future parallel pipelines
- Crossing rivers and canals below grade
- Supporting pipe in unstable sediments
- Relieving pressure during rapid flow changes

Historical encounters

1. **Acquire a right-of-way wide enough for four eventual pipelines.**
2. **Organize distributed welding, riveting and pipe-laying crews.**
3. **Carry the aqueduct beneath the San Joaquin River and Elliott Cut.**

Phase 5: Coast Range Tunnel

Game miles 97–126 | Primarily 1927–1934

Outcome: The player confronts the most dangerous and financially uncertain tunneling division of the project.

The Coast Range Tunnel extended approximately 28.5 miles from Tesla toward Alameda Creek and Irvington. The work encountered groundwater, gas and ground that deformed after excavation.

Principal physical challenges

- Choosing a gravity tunnel over a lower-capacity pumped alternative
- Deep shafts and long underground headings
- Swelling and squeezing ground
- Methane and ventilation hazards
- Maintaining safety after a fatal explosion
- Funding interruptions during the Depression
- Preserving alignment between isolated headings

Historical encounters

1. **Commit to a gravity tunnel rather than permanent pumping.**
2. **Develop a new support method at Crane Ridge.**
3. **Respond to the Mitchell Shaft explosion.**
4. **Secure bond funding, restart suspended headings and complete the final breakthrough.**

Phase 6: Sunol, Bay Crossing and Pulgas

Game miles 126–167 | Primarily 1922–1934

Outcome: Previously completed western infrastructure must be connected to the advancing Hetch Hetchy system and placed into service as one continuous water-supply chain.

West of the Coast Range Tunnel, water crossed Alameda Creek through an inverted siphon, continued through the Irvington area, crossed the southern portion of San Francisco Bay and reached Pulgas. Another approximately 25 miles of conveyance extended west from Irvington toward the Peninsula.

Principal physical challenges

- Crossing Alameda Creek and the Sunol Valley
- Financing construction when city bond funds were unavailable
- Burying a large pipeline in deep bay mud
- Preserving a trained workforce during funding interruptions
- Connecting independently completed facilities
- Coordinating the first controlled arrival of Sierra water

Historical encounters

1. **Complete the Alameda Creek inverted siphon.**
2. **Finance and construct the first Bay Crossing pipeline.**
3. **Connect the route at Pulgas and deliver the first Hetch Hetchy water.**

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5. Final Event-Card Table

The resource effects below are recommended game-balancing values. They are not historical measurements.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
1. Hetch Hetchy and Canyon Ranch	Road work began around Hog Ranch in 1914. A sawmill at Canyon Ranch began operating in July 1915, approximately 4.5 miles from the dam site. The sawmill and its later replacement at Hog Ranch produced millions of board feet of construction lumber.	**Cut the First Road.** Heavy work cannot begin until the player builds mountain access and establishes a local timber supply. Investing fully makes later dam and railroad cards faster and improves camp conditions.	**Full build:** Funds −2, Crew +1, Time −1 on two later High Sierra cards. **Minimal access:** Funds −1 now, Crew −1 and Time +1 later.
1. Hetch Hetchy Railroad	The city constructed a 68-mile standard-gauge railroad from Hetch Hetchy Junction to the mountain project. It was completed in 1917 and carried cement, equipment, workers and supplies. It operated in winter when mountain roads could be blocked by snow.	**Build the Railroad.** Spend heavily now to remove recurring freight delays. During winter, an operational railroad prevents a seasonal shutdown.	Funds −3, Time −2 over the phase, Crew +1. Ignore one future snow-related delay.
1. Railroad Excursions and Bond Support	San Francisco residents were taken on project excursions, helping residents see the work and strengthening public interest in the water project and its financing.	**Bring the Public to the Project.** Dedicate scarce passenger capacity to a public inspection tour or reserve every train for freight.	**Run excursion:** Funds −1, Public Support +2, Time +1. **Freight only:** Time −1, but no support gain.
1. O'Shaughnessy Dam Site	Work required a roughly 1,000-foot diversion tunnel and deep excavation through glacial boulders, sand and other material before sound bedrock could be reached. The initial dam was completed in 1923 after continuous, all-season work.	**Find Bedrock.** Test borings reveal that the foundation excavation will be much deeper than the visible valley floor suggests. The player must fund full excavation and river diversion before concrete placement can continue.	Funds −3, Time +2, Crew −1, Water Readiness +3 when complete.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
2. Early Intake	Construction of Early Intake Powerhouse began in 1917. Generation began in May 1918, supplying electricity east to the dam site and west toward Moccasin. Surplus electricity was also sold commercially.	**Power Before Water.** Build local generation before expanding round-the-clock drilling. The initial delay produces faster work and limited revenue during later construction.	Funds −2, Time +1 now, then Time −2 across Mountain Tunnel cards; Public Support +1; Water Readiness +1.
2. Mountain Tunnel	Mountain Tunnel was excavated from 12 faces. Access included the 786-foot Second Garrote shaft and 646-foot Big Creek shaft. The tunnel was completed in 1925.	**Twelve Faces, One Line.** Open additional shafts and headings to attack the tunnel simultaneously. More faces accelerate excavation but require more equipment, surveying and supervision.	**All headings:** Funds −3, Time −2, Crew −1, Water Readiness +2. **Reduced headings:** Funds −1, Time +2, Crew +1.
2. Priest Reservoir and Moccasin	Priest Reservoir received Mountain Tunnel water. Rattlesnake Creek was diverted to reduce contamination risk. Water then descended approximately 1,316 feet through penstocks to Moccasin Powerhouse. Mountain Tunnel water reached Priest in June 1925, and Moccasin entered commercial operation that August.	**Protect the Reservoir, Harness the Drop.** Complete the creek diversion, reservoir and penstocks as one coordinated system. This costs time but improves water quality and turns elevation into useful power.	Funds −2, Time +1, Water Readiness +2, Public Support +1.
3. Foothill Tunnel Camps	Six construction camps supported Foothill Tunnel work. City crews installed camp water, electricity, telephones, roads and other support facilities before and during excavation.	**Six Camps in the Foothills.** Decide how much infrastructure to provide before distributing the tunneling crews. Well-supplied camps retain workers and reduce future interruptions.	**Fully serviced camps:** Funds −2, Crew +2, Time −1 on this phase. **Bare camps:** Funds −1, Crew −2, Time +1.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
3. Red Mountain Bar	The aqueduct crossed Red Mountain Bar through a 770-foot steel inverted siphon anchored in bedrock and concrete. The crossing was necessary because the canyon would be affected by the filling of Don Pedro Reservoir. A long cableway transported workers and loads across the canyon.	**Beat the Rising Reservoir.** Complete the cableway and lower the siphon sections before access to the canyon becomes more difficult.	Funds −2, Time −1, Crew −1, Water Readiness +1. A delay adds Time +2 and Funds −1.
3. Hetch Hetchy Junction Heading	City crews and contractors competed for tunneling progress. A heading advanced 803 feet during September 1926, following another 781-foot month earlier that year. The Foothill Tunnel was completed in 1929.	**The 803-Foot Month.** Management challenges the crew to pursue another excavation record. The player chooses between maximum advance and a sustainable production schedule.	**Record pace:** Time −2, Funds +1, Crew −2. **Sustainable pace:** Funds −1, Crew +2, no time change.
4. Oakdale-to-Tesla Right-of-Way	The city obtained a right-of-way approximately 100 feet wide, providing room for as many as four parallel San Joaquin pipelines.	**Leave Room for Four Pipes.** Acquire more land than Pipeline No. 1 immediately requires. The larger purchase reduces current funds but creates future expansion capacity.	Funds −2, Water Readiness +1 and permanent **Expansion Capacity +2**. A narrow purchase saves Funds +1 but removes the expansion bonus.
4. San Joaquin Pipeline No. 1	The first San Joaquin pipeline extended approximately 47.5 miles between Oakdale and Tesla. Built in 1931–1932, it used welded and riveted steel sections ranging from about 56 to 72 inches in diameter.	**Forty-Seven Miles of Steel.** Divide crews among pipe delivery, trench preparation, welding, riveting and inspection. Distributed work is faster but increases coordination and fatigue.	Funds −2, Time −2, Crew −1, Water Readiness +2.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
4. San Joaquin River and Elliott Cut	Portions of the pipelines descended below sea level and passed beneath the San Joaquin River and Elliott Cut. Pipes were supported on timber piles and protected with concrete jackets. Relief valves were included to control excess pressure.	**Under the San Joaquin.** Wait for favorable conditions, install deep foundations and include pressure relief equipment rather than treating the valley as a simple surface crossing.	Funds −2, Time +1, Crew +1, Water Readiness +2. Skipping the favorable season causes Crew −2 and Time +1.
5. Tesla Portal and Coast Range	Construction of the 28.5-mile Coast Range Tunnel was delayed until 1927. A pumped pipeline was discussed as an alternative, but the gravity tunnel offered greater capacity without permanent pumping costs.	**Gravity or Pumps?** Make the project's defining long-term decision. The gravity tunnel is slower and more expensive to construct, while the debated pumped alternative would provide less capacity and continuing operating costs. Only the gravity option follows the completed historical system.	**Gravity tunnel:** Funds −3, Time +2, Water Readiness +3 and no recurring pumping charge. **Pumped alternative:** Funds −2, Time −1, Water Readiness +1 and Funds −1 during every later operating chapter.
5. Crane Ridge	At Crane Ridge, approximately 2,500 feet below the surface, swelling ground squeezed the excavated opening. In one reported section, an approximately 18-foot opening contracted to about three feet within 24 hours. Crews responded by excavating an oversized bore and constructing gunite support rings with room for the ground to move.	**Crane Ridge Closes In.** Previously reliable timber supports begin crushing. Pause excavation and adopt the larger gunite-ring design.	Funds −2, Time +1, Crew +1, Water Readiness +2. Continuing with the original support method causes Time +2 and Crew −2.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
5. Mitchell Shaft	Methane was encountered during Coast Range Tunnel construction. On July 17, 1931, an explosion at Mitchell Shaft killed 12 workers. Investigations documented existing safety precautions but did not establish one conclusive cause.	**Mitchell Shaft Memorial.** This should be a fixed historical event, not a random or easily preventable player failure. Stop work, account for the dead, support the crews, review ventilation and reinforce gas monitoring before excavation resumes.	Immediate Crew −3, Time +2 and Public Support −1. **Full safety stand-down:** Funds −1, Crew +1, Public Support +1 before restart.
5. Coast Range Headings	Financial shortages temporarily suspended portions of the tunnel work. A 1932 bond issue provided additional money, and the last Coast Range Tunnel connection was made on January 5, 1934, between the Mitchell and Mocho headings.	**Sell the Bonds, Finish the Bore.** Active headings are running out of authorized funding. Conduct a public bond campaign, restart the crews and preserve enough surveying precision for the final breakthrough.	On passage: Funds +3, Public Support +1, Time −2 and Water Readiness +3. On failure: Time +2 and Crew −1 before the vote can be attempted again.
6. Alameda Creek and Sunol Valley	The Coast Range system connected to Irvington through a multiple-pipe inverted siphon across Alameda Creek and the Sunol area. Alameda Creek Siphon No. 1 was completed in February 1934.	**Across Sunol Valley.** Install the siphon sections and coordinate the crossing with completion of the Coast Range headings.	Funds −2, Time +1, Crew −1, Water Readiness +2. Completing it before the tunnel is ready allows one crew to be reassigned, producing Crew +1 later.

Phase/Location	Historical Fact	Game Event Description	Impacted Resources
6. Newark Slough and Bay Crossing	When city funds were exhausted, the Spring Valley Water Company advanced financing that helped prevent a prolonged shutdown and layoffs. Bay Crossing Pipeline No. 1 was placed in a trench through bay mud, reaching approximately 75 feet below the water surface in places. It was completed in 1925 and immediately used for local water transfers.	**Mud, Bay and Bridge Finance.** Accept interim financing to retain hundreds of trained workers, then bury the large steel pipeline through the southern bay crossing.	**Accept advance:** Funds +2, Time −2, Crew +1, Public Support −1. **Construction:** Funds −2, Crew −1, Water Readiness +2.
6. Pulgas and Peninsula Connection	Pulgas Tunnel was begun in 1922 and carried Spring Valley water before the full Hetch Hetchy route was complete. The first Hetch Hetchy water reached Pulgas at 10:12 a.m. on October 24, 1934. A major public celebration followed on October 28.	**The First Drop at Pulgas.** Confirm that every upstream division is connected, open the system and convert accumulated Water Readiness into delivered water. Hold a public ceremony recognizing the workforce and the completed gravity system.	Water Supply +5, Public Support +3, Crew +1, Time status becomes **System Complete**. Optional ceremony: Funds −1.

6. Chronological Construction Milestones

Use this list for chapter unlocks, loading-screen dates or campaign milestones.

Year	Milestone
1914	Road and railroad-route work begins in the High Sierra project area.
1915	Canyon Ranch sawmill begins operating; river-diversion and railroad construction preparations advance.
1917	Hetch Hetchy Railroad is completed; work begins on Early Intake power facilities and Mountain Tunnel.
1918	Early Intake begins generating construction power and selling surplus electricity.
1919	Major O’Shaughnessy Dam construction begins.

Year	Milestone
1922	Work begins on Pulgas Tunnel while major eastern components remain under construction.
1923	Initial O'Shaughnessy Dam and Priest Dam are completed; Hetch Hetchy Reservoir first fills.
1924	Pulgas Tunnel carries Spring Valley water, demonstrating that western facilities can operate before Sierra water arrives.
1925	Mountain Tunnel reaches Priest; Moccasin Powerhouse enters operation; Bay Crossing Pipeline No. 1 is completed.
1926	Foothill Tunnel excavation expands, including record monthly advances.
1927	Major Coast Range Tunnel work begins after funding delay.
1929	Foothill Tunnel is completed.
1931	San Joaquin Pipeline work advances; the Mitchell Shaft explosion kills 12 workers on July 17.
1932	San Joaquin Pipeline No. 1 is completed and new bond authority supports continuation of Coast Range work.
January 1934	The last Coast Range Tunnel headings connect.
February 1934	Alameda Creek Siphon No. 1 is completed.
October 24, 1934	Hetch Hetchy water reaches Pulgas.
October 28, 1934	The completed water system is publicly celebrated.

7. Important Visual-History Note

For a 1934 completion scene, depict the **temporary ceremonial structure** erected for the first-water celebration rather than the permanent Pulgas Water Temple commonly photographed today. The permanent temple was completed in 1938.

8. Recommended Godot Custom Resource Fields

Each encounter can be stored with the following fields:

```
event_id: StringName
event_title: String
phase_id: StringName

mile_start: int
mile_end: int
historical_year_start: int
historical_year_end: int

location_name: String
historical_fact: String
event_description: String

funds_delta: int
public_support_delta: int
water_readiness_delta: int
crew_wellbeing_delta: int
time_delta_seasons: int

required_flags: Array[StringName]
granted_flags: Array[StringName]

canonical_choice: int
historical_source_note: String
assumption_note: String
```

Useful campaign flags

- `high_sierra_access_complete`
- `railroad_operational`
- `construction_power_available`
- `dam_complete`
- `mountain_tunnel_complete`
- `moccasin_power_available`
- `foothill_tunnel_complete`
- `san_joaquin_pipeline_complete`
- `coast_range_tunnel_complete`
- `alameda_siphon_complete`
- `bay_crossing_complete`
- `pulgas_connected`
- `hetch_hetchy_water_delivered`

Recommended completion rule

Do not convert Water Readiness into delivered Water Supply until all mandatory route flags are true:

```python
system_connected = (
    dam_complete
    and mountain_tunnel_complete
    and foothill_tunnel_complete
    and san_joaquin_pipeline_complete
    and coast_range_tunnel_complete
    and alameda_siphon_complete
    and bay_crossing_complete
    and pulgas_connected
)
```

This makes the final Pulgas encounter mechanically meaningful: every individual structure may be complete, but the city receives no Hetch Hetchy water until the entire chain functions as one system.